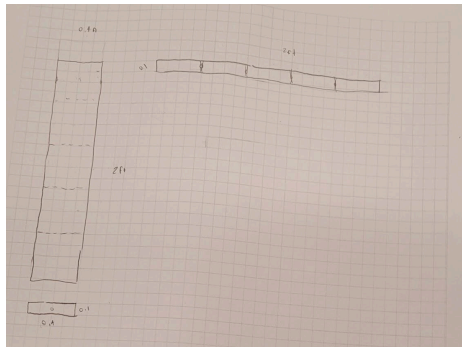


Element D:

Solution A is a 3D printed “H” shaped piece that slides in between the doors slit and sits upon the lock. This design is small, lightweight, and effective. This will be 3D printed with PLA filament.

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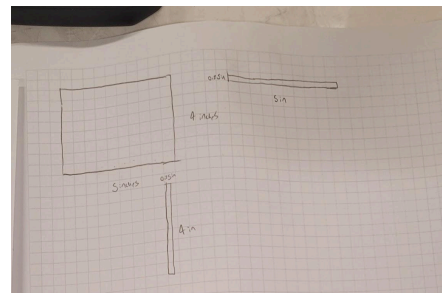


Solution B is a collapsable arm for reaching over the door, to slide the lock closed from the outside. This would also be 3d printed with PLA and would collapse to be able to be put in the user’s pocket. It would consist of multiple different pieces that click together

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Solution C is the most simple. Solution C is just 2 large thin magnets that you place on each side, slipping one through the door.

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Decision Matrix

		<u>Solution A</u>	<u>Solution A</u>	<u>Solution B</u>	<u>Solution B</u>	<u>Solution C</u>	<u>Solution C</u>
<u>Criteria</u>	<u>Weight</u>	<u>Rating</u>	<u>Weight Score</u>	<u>Rating</u>	<u>Weight Score</u>	<u>Rating</u>	<u>Weight Score</u>
<u>Closure</u>	<u>35%</u>	<u>3</u>	<u>1.05</u>	<u>1</u>	<u>0.35</u>	<u>2</u>	<u>0.7</u>
<u>Durability</u>	<u>15%</u>	<u>2</u>	<u>0.3</u>	<u>1</u>	<u>0.15</u>	<u>3</u>	<u>0.45</u>
<u>Endurance</u>	<u>15%</u>	<u>3</u>	<u>0.45</u>	<u>2</u>	<u>0.3</u>	<u>1</u>	<u>0.15</u>
<u>Weight</u>	<u>15%</u>	<u>3</u>	<u>0.45</u>	<u>2</u>	<u>0.3</u>	<u>1</u>	<u>0.15</u>
<u>Texture</u>	<u>10%</u>	<u>2</u>	<u>0.2</u>	<u>1</u>	<u>0.1</u>	<u>3</u>	<u>0.3</u>
<u>Speed</u>	<u>10%</u>	<u>3</u>	<u>0.3</u>	<u>1</u>	<u>0.1</u>	<u>2</u>	<u>0.2</u>
<u>Price</u>	<u>5%</u>	<u>2</u>	<u>0.1</u>	<u>1</u>	<u>0.05</u>	<u>3</u>	<u>0.15</u>
<u>Overall Score</u>			<u>2.85</u>		<u>1.35</u>		<u>2.1</u>

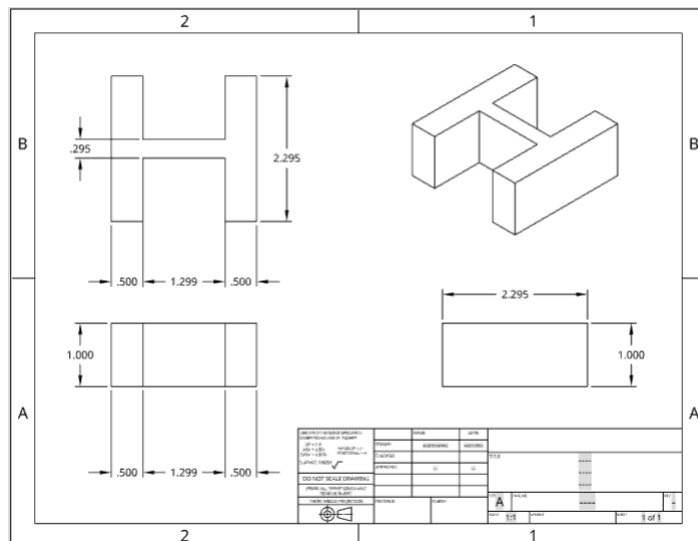
Solution A is the best.

Design Blueprint

CAD link:

<https://cad.onshape.com/documents/ef8983a79edeae6a689be463/w/fdc0f4fcf8bf682312f1ca1e/e/7941be0a486be3d75260bc7c?renderMode=0&uiState=69825ab10554a1c5086ca062>

1/21/26



Manufacturing Method: 3D-Printing

Steps:

1. Export the 3D design from Onshape as an stl file
2. Open the file in Bambu Lab Studio
3. Set up the respective 3D printer with your desired settings
4. Print the model
5. Wait for 10 minutes after the print is finished to take it off the build plate.
6. Done

We have to use items that do not have any cost, since the child's parent requires that we use materials with zero cost.